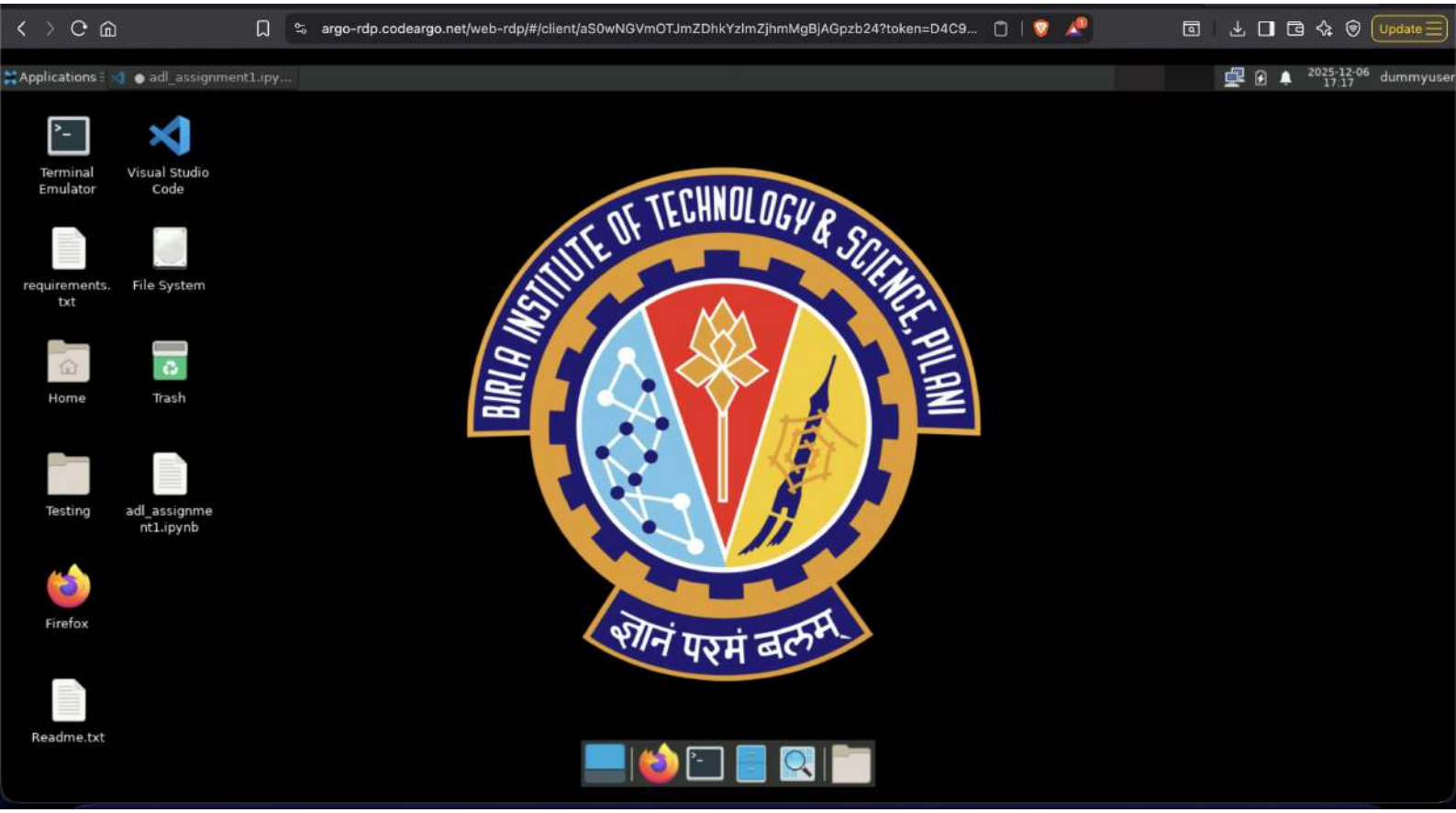
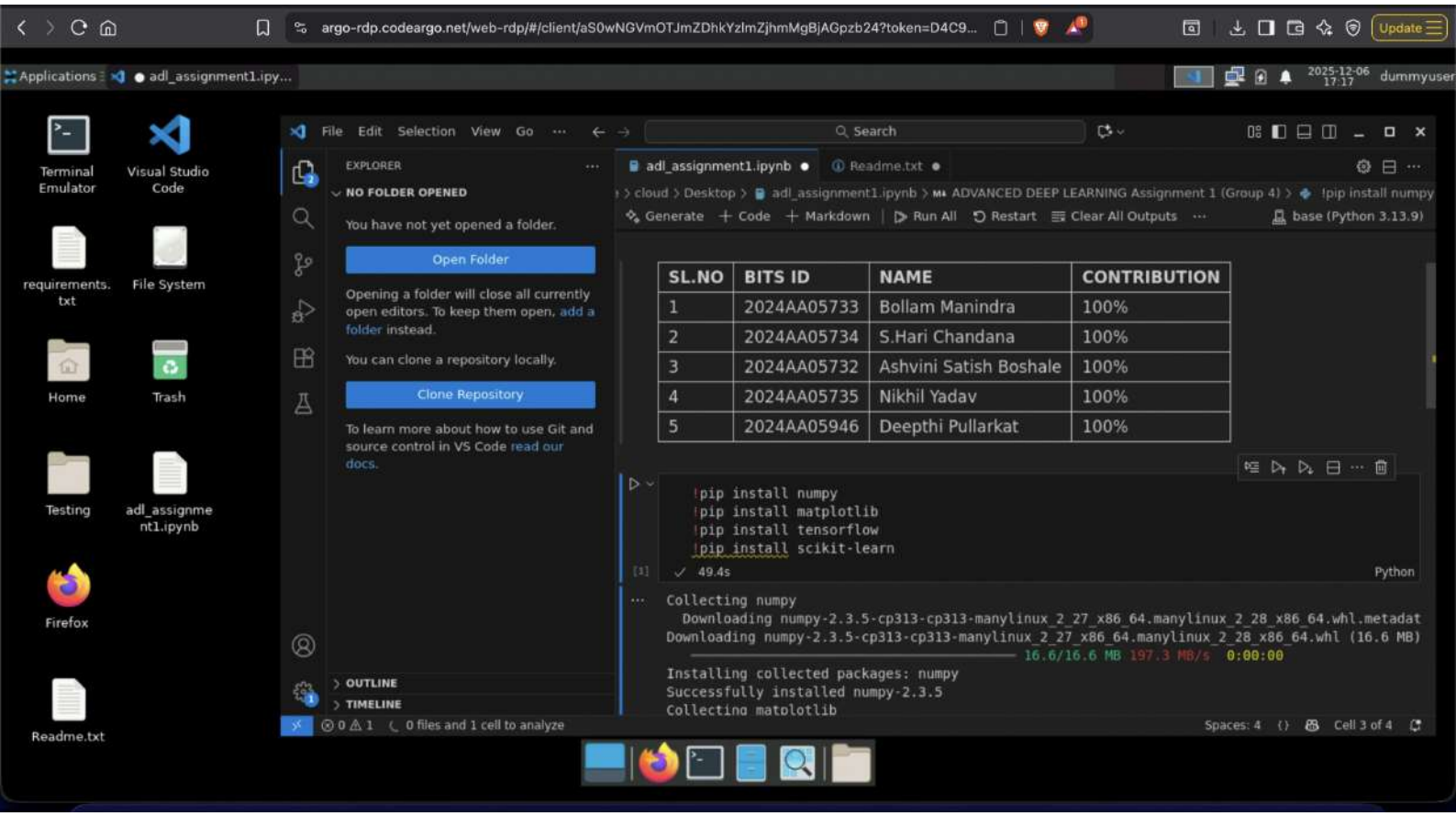
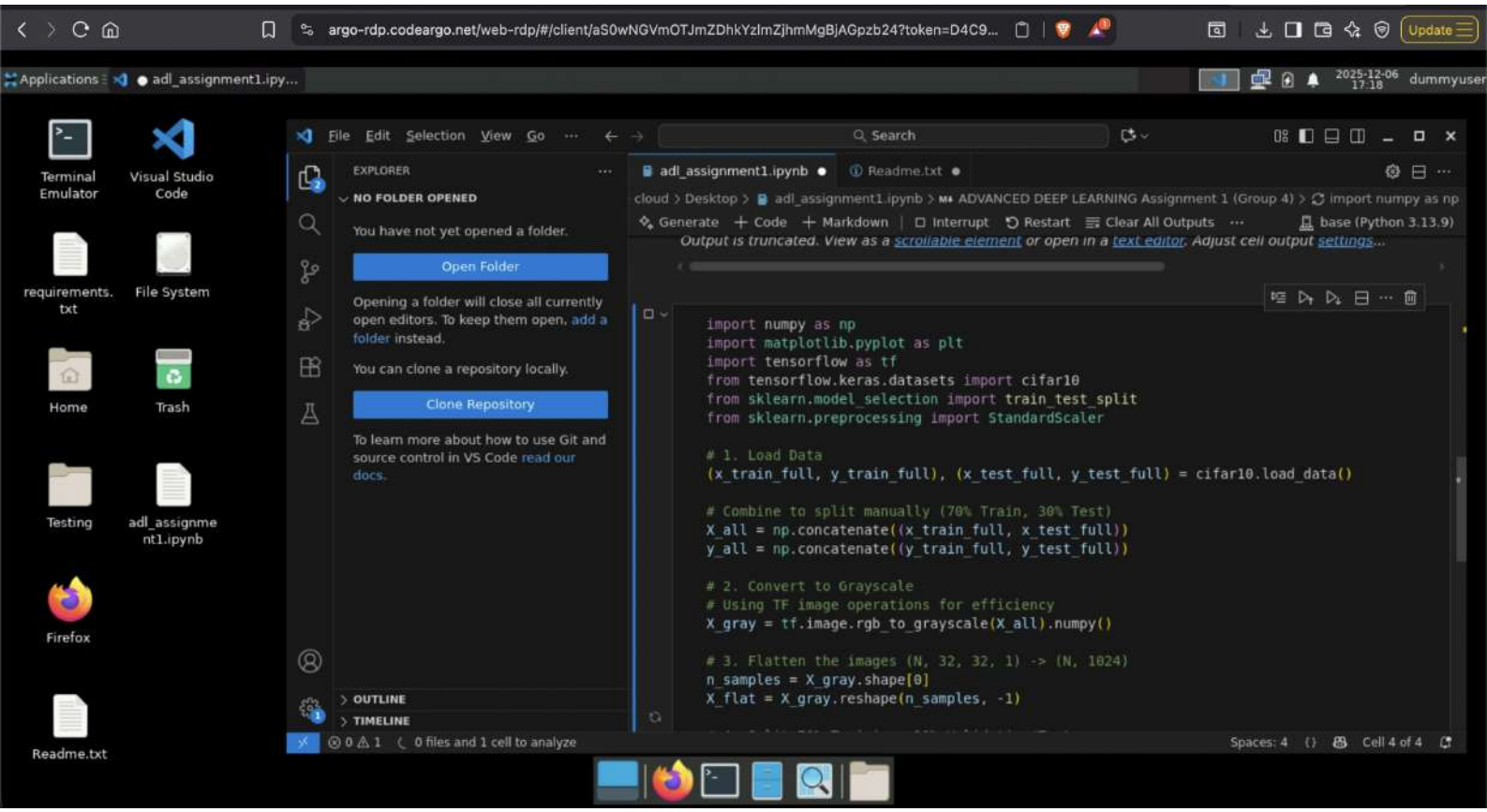
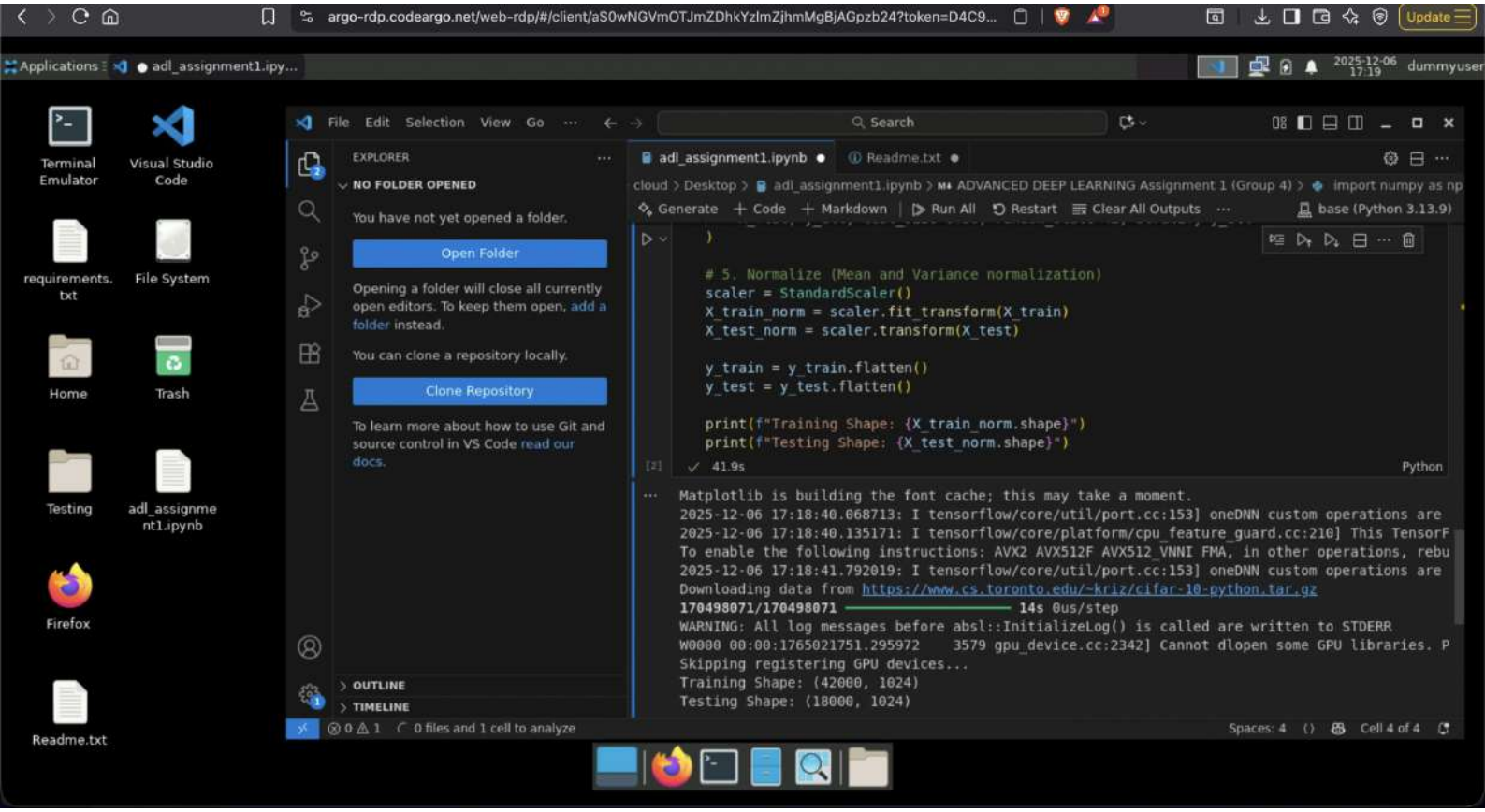
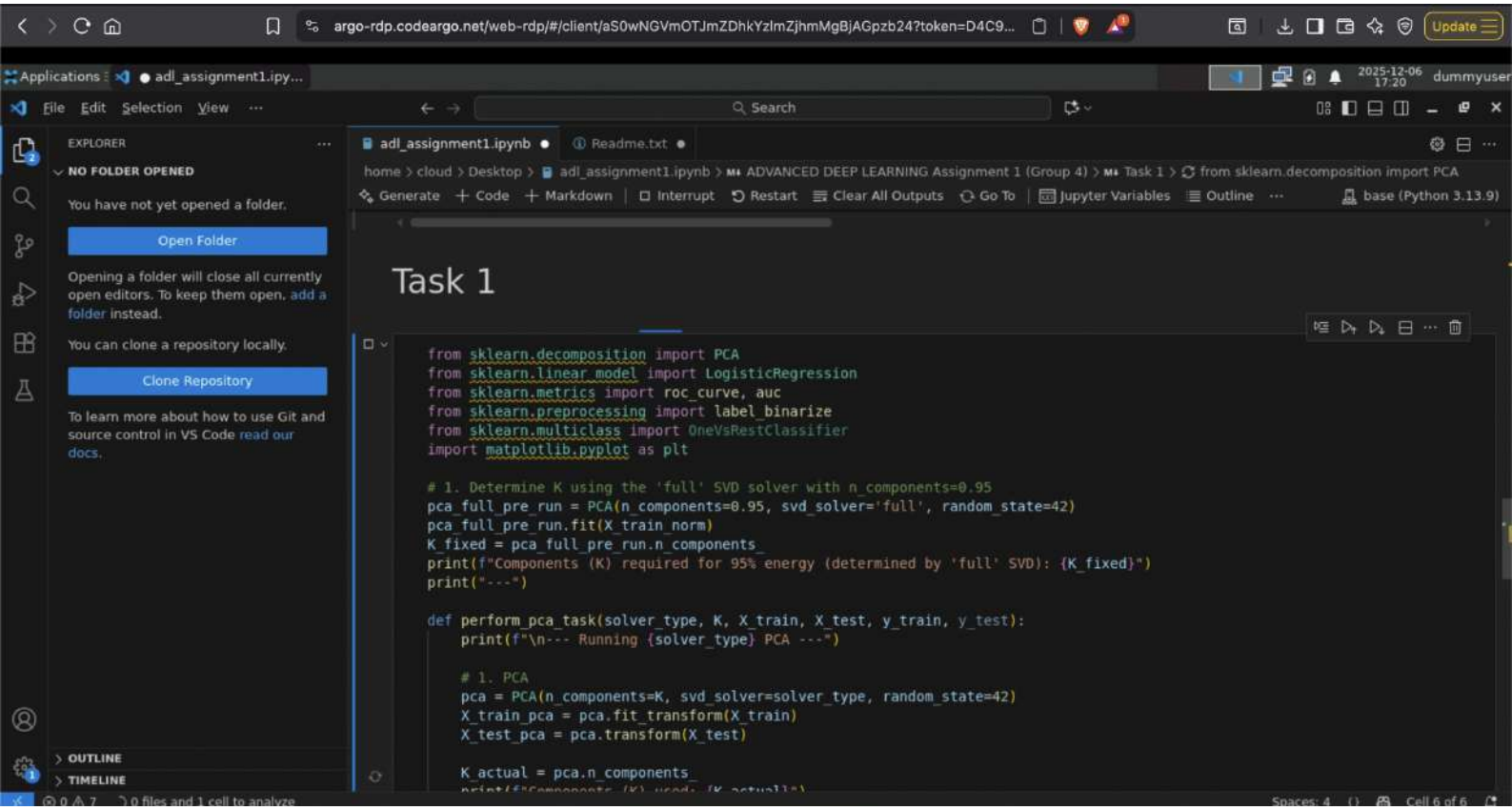

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Terminal Emulator
Visual Studio Code
requirements.txt
File System
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Trash
Testing
adl_assignment1.ipynb
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Readme.txt
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2025-12-06 17:17
dummyuser
Terminal Emulator
Visual Studio Code
requirements.txt
File System
Home
Trash
Testing
adl_assignment1.ipynb
Firefox
Readme.txt
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OUTLINE
TIMELINE
adl_assignment1.ipynb
Readme.txt
cloud > Desktop > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > !pip install numpy
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
base (Python 3.13.9)
SL.NO | BITS ID | NAME | CONTRIBUTION
1 | 2024AA05733 | Bollam Manindra | 100%
2 | 2024AA05734 | S.Hari Chandana | 100%
3 | 2024AA05732 | Ashvini Satish Boshale | 100%
4 | 2024AA05735 | Nikhil Yadav | 100%
5 | 2024AA05946 | Deepthi Pullarkat | 100%
!pip install numpy
!pip install matplotlib
!pip install tensorflow
!pip install scikit-learn
[3]
49.4s
Python
Collecting numpy
Downloading numpy-2.3.5-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadat
Downloading numpy-2.3.5-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.6 MB)
16.6/16.6 MB 197.3 MB/s 0:00:00
Installing collected packages: numpy
Successfully installed numpy-2.3.5
Collecting matplotlib
0 files and 1 cell to analyze
Spaces: 4
Cell 3 of 4


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Terminal Emulator
Visual Studio Code
requirements.txt
File System
Home
Trash
Testing
adl_assignment1.ipynb
Firefox
Readme.txt
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Search
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OUTLINE
TIMELINE
adl_assignment1.ipynb
Readme.txt
cloud > Desktop > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > import numpy as np
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Clear All Outputs
base (Python 3.13.9)
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.datasets import cifar10
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
1. Load Data
(x_train_full, y_train_full), (x_test_full, y_test_full) = cifar10.load_data()
Combine to split manually (70% Train, 30% Test)
X_all = np.concatenate((x_train_full, x_test_full))
y_all = np.concatenate((y_train_full, y_test_full))
2. Convert to Grayscale
Using TF image operations for efficiency
X_gray = tf.image.rgb_to_grayscale(X_all).numpy()
3. Flatten the images (N, 32, 32, 1) -> (N, 1024)
n_samples = X_gray.shape[0]
X_flat = X_gray.reshape(n_samples, -1)
0 files and 1 cell to analyze
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Cell 4 of 4


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dummyuser
Terminal Emulator
Visual Studio Code
requirements.txt
File System
Home
Trash
Testing
adl_assignment1.ipynb
Firefox
Readme.txt
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OUTLINE
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Readme.txt
cloud > Desktop > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > import numpy as np
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
base (Python 3.13.9)
5. Normalize (Mean and Variance normalization)
scaler = StandardScaler()
X_train_norm = scaler.fit_transform(X_train)
X_test_norm = scaler.transform(X_test)
y_train = y_train.flatten()
y_test = y_test.flatten()
print(f"Training Shape: {X_train_norm.shape}")
print(f"Testing Shape: {X_test_norm.shape}")
41.9s
Python
Matplotlib is building the font cache; this may take a moment.
2025-12-06 17:18:40.068713: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
2025-12-06 17:18:40.135171: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorF
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operations, rebu
2025-12-06 17:18:41.792019: I tensorflow/core/util/port.cc:153] oneDNN custom operations are
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 14s 0us/step
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
W0000 00:00:1765021751.295972 3579 gpu_device.cc:2342] Cannot dlopen some GPU libraries. P
Skipping registering GPU devices...
Training Shape: (42000, 1024)
Testing Shape: (18000, 1024)
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Cell 4 of 4


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adl_assignment1.ipynb
Readme.txt
home > cloud > Desktop > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > Task 1 > from sklearn.decomposition import PCA
Generate Code Markdown Interrupt Restart Clear All Outputs Go To Jupyter Variables Outline
base (Python 3.13.9)
Task 1
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize
from sklearn.multiclass import OneVsRestClassifier
import matplotlib.pyplot as plt
1. Determine K using the 'full' SVD solver with n_components=0.95
pca_full_pre_run = PCA(n_components=0.95, svd_solver='full', random_state=42)
pca_full_pre_run.fit(X_train_norm)
K_fixed = pca_full_pre_run.n_components_
print(f"Components (K) required for 95% energy (determined by 'full' SVD): {K_fixed}")
print("---")
def perform_pca_task(solver_type, K, X_train, X_test, y_train, y_test):
print(f"\n--- Running {solver_type} PCA ---")
1. PCA
pca = PCA(n_components=K, svd_solver=solver_type, random_state=42)
X_train_pca = pca.fit_transform(X_train)
X_test_pca = pca.transform(X_test)
K_actual = pca.n_components_
OUTLINE
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Cell 6 of 6

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OUTLINE

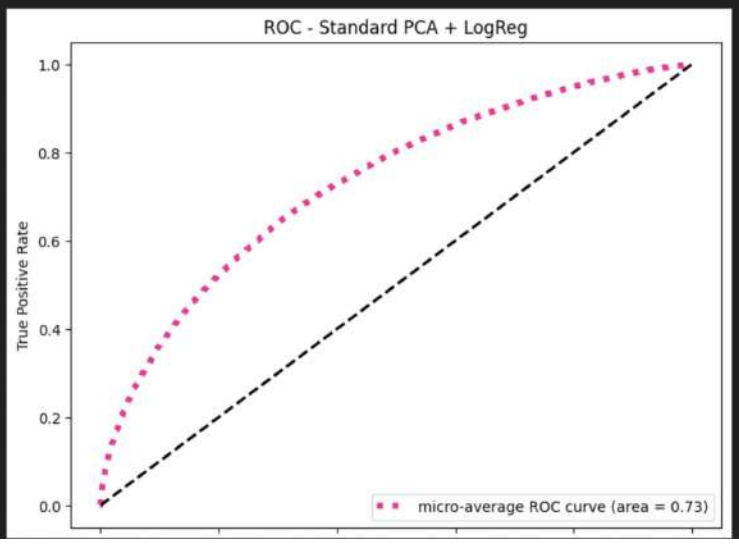
TIMELINE

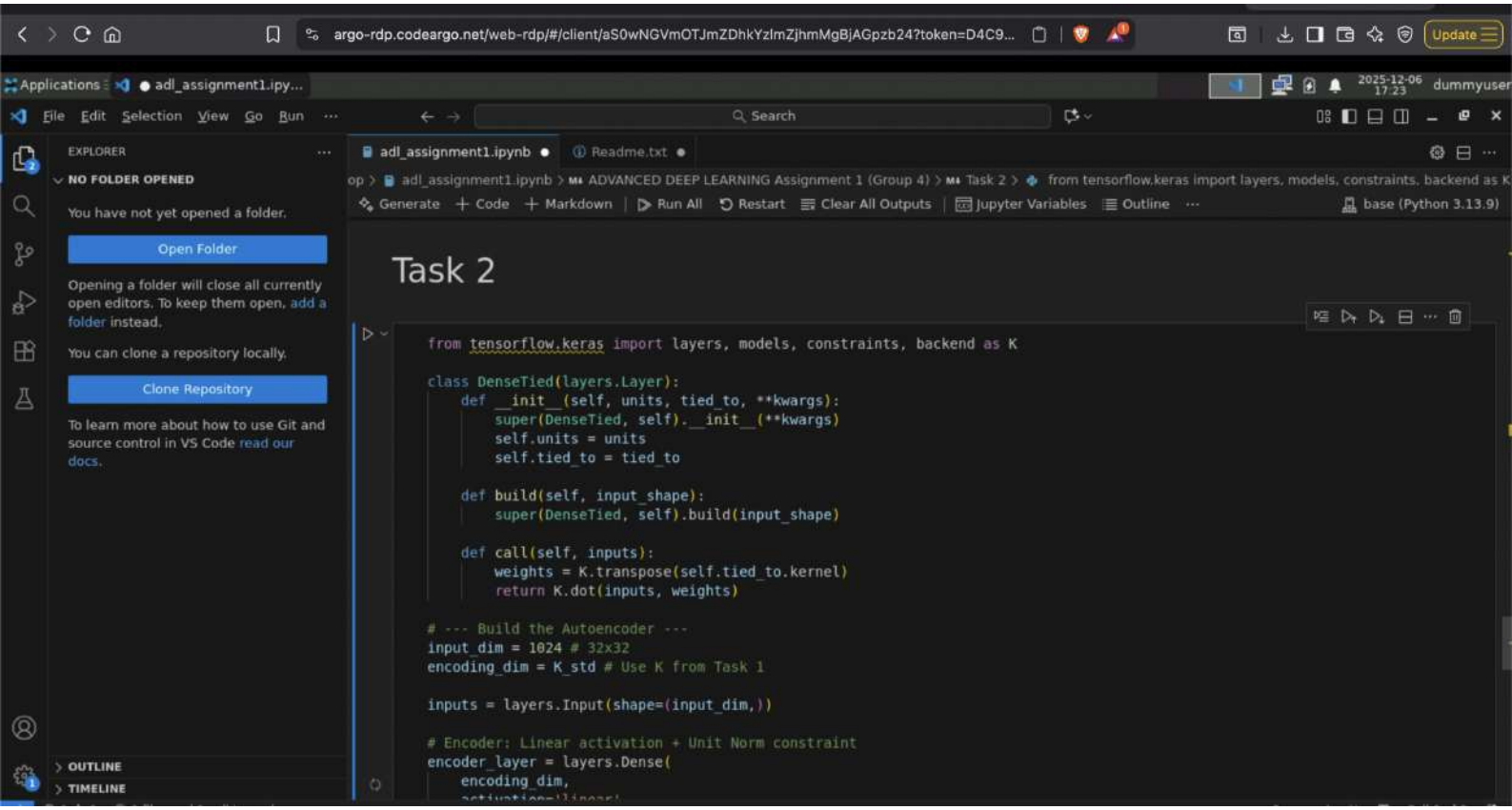
adl_assignment1.ipynb ● Readme.txt ●

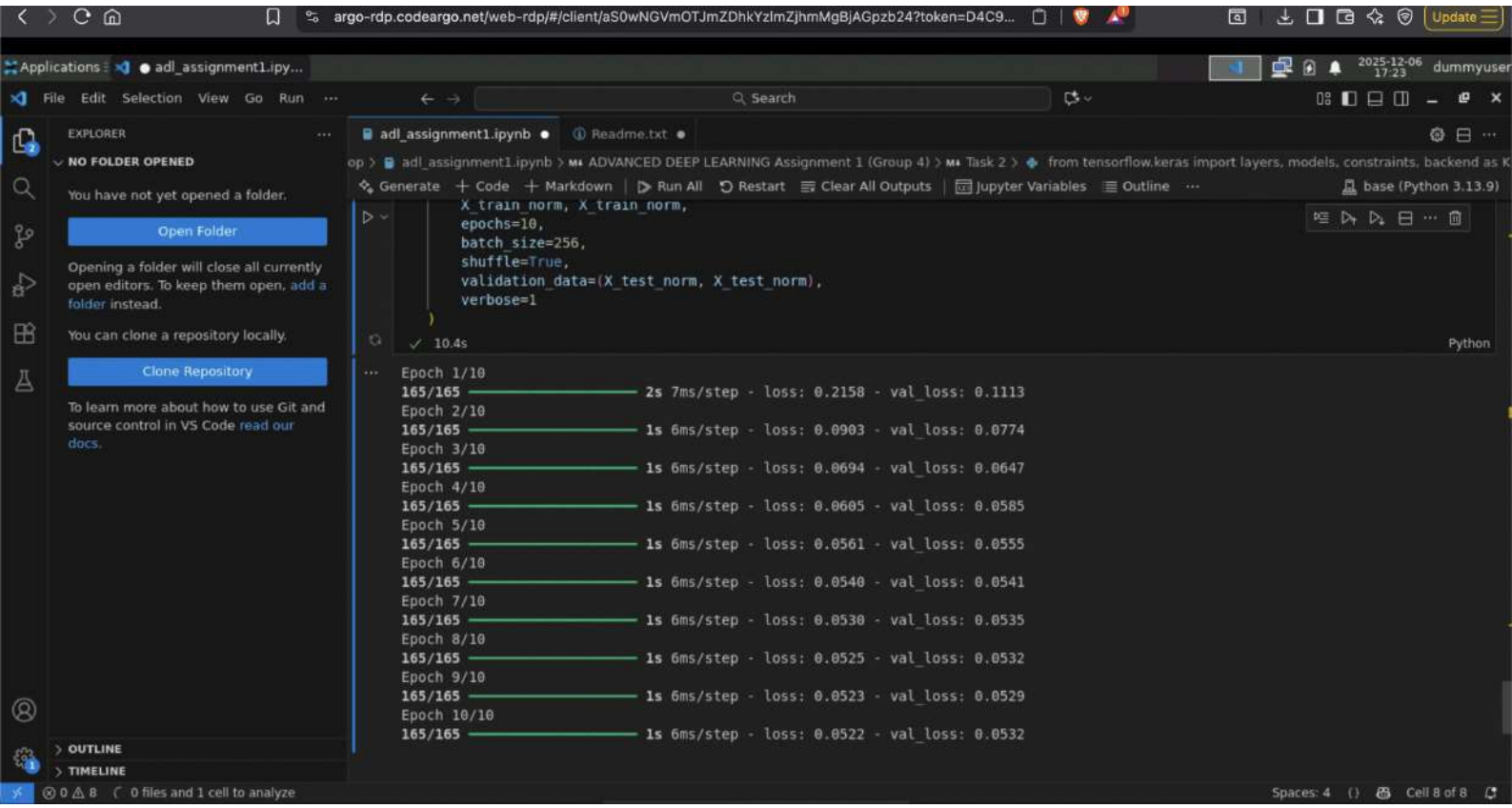
home > cloud > Desktop > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > Task 1 > from sklearn.decomposition import PCA

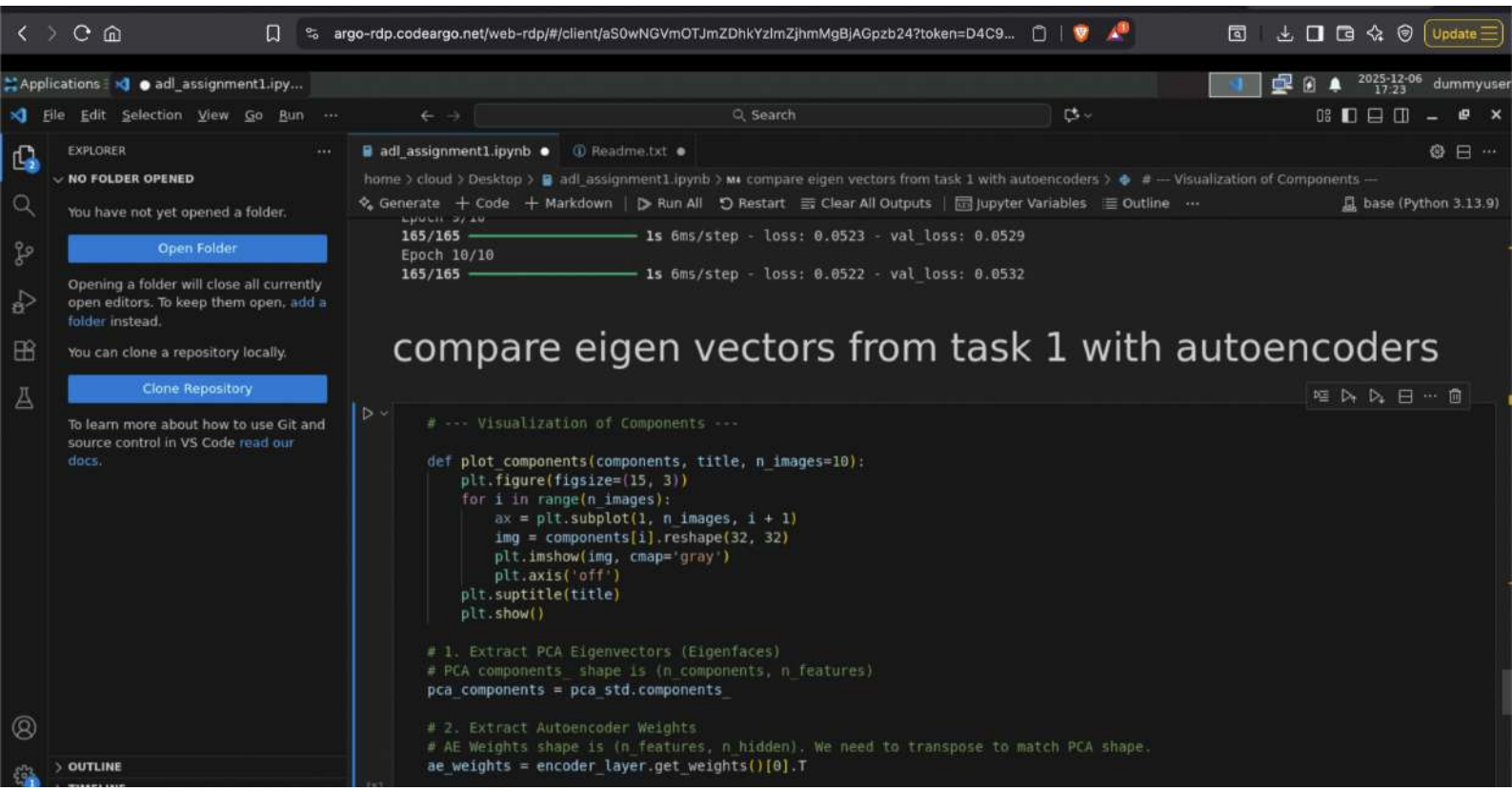
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ··· base (Python 3.13.9)

```
--- Running randomized PCA ---
Components (K) used: 163
```




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OUTLINE
TIMELINE
adl_assignment1.ipynb
Readme.txt
op > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > Task 2 > from tensorflow.keras import layers, models, constraints, backend as K
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
base (Python 3.13.9)
Task 2
from tensorflow.keras import layers, models, constraints, backend as K
class DenseTied(layers.Layer):
def __init__(self, units, tied_to, **kwargs):
super(DenseTied, self).__init__(**kwargs)
self.units = units
self.tied_to = tied_to
def build(self, input_shape):
super(DenseTied, self).build(input_shape)
def call(self, inputs):
weights = K.transpose(self.tied_to.kernel)
return K.dot(inputs, weights)
--- Build the Autoencoder ---
input_dim = 1024 # 32x32
encoding_dim = K_std # Use K from Task 1
inputs = layers.Input(shape=(input_dim,))
Encoder: Linear activation + Unit Norm constraint
encoder_layer = layers.Dense(
encoding_dim,


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op > adl_assignment1.ipynb > ADVANCED DEEP LEARNING Assignment 1 (Group 4) > Task 2 > from tensorflow.keras import layers, models, constraints, backend as K
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
base (Python 3.13.9)
X_train_norm, X_train_norm,
epochs=10,
batch_size=256,
shuffle=True,
validation_data=(X_test_norm, X_test_norm),
verbose=1
)
10.4s
Python
Epoch 1/10
165/165 2s 7ms/step - loss: 0.2158 - val_loss: 0.1113
Epoch 2/10
165/165 1s 6ms/step - loss: 0.0903 - val_loss: 0.0774
Epoch 3/10
165/165 1s 6ms/step - loss: 0.0694 - val_loss: 0.0647
Epoch 4/10
165/165 1s 6ms/step - loss: 0.0605 - val_loss: 0.0585
Epoch 5/10
165/165 1s 6ms/step - loss: 0.0561 - val_loss: 0.0555
Epoch 6/10
165/165 1s 6ms/step - loss: 0.0540 - val_loss: 0.0541
Epoch 7/10
165/165 1s 6ms/step - loss: 0.0530 - val_loss: 0.0535
Epoch 8/10
165/165 1s 6ms/step - loss: 0.0525 - val_loss: 0.0532
Epoch 9/10
165/165 1s 6ms/step - loss: 0.0523 - val_loss: 0.0529
Epoch 10/10
165/165 1s 6ms/step - loss: 0.0522 - val_loss: 0.0532
0 8
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Cell 8 of 8


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Search
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OUTLINE
adl_assignment1.ipynb
Readme.txt
home > cloud > Desktop > adl_assignment1.ipynb > compare eigen vectors from task 1 with autoencoders > # --- Visualization of Components ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
base (Python 3.13.9)
165/165 1s 6ms/step - loss: 0.0523 - val_loss: 0.0529
Epoch 10/10
165/165 1s 6ms/step - loss: 0.0522 - val_loss: 0.0532
compare eigen vectors from task 1 with autoencoders
--- Visualization of Components ---
def plot_components(components, title, n_images=10):
plt.figure(figsize=(15, 3))
for i in range(n_images):
ax = plt.subplot(1, n_images, i + 1)
img = components[i].reshape(32, 32)
plt.imshow(img, cmap='gray')
plt.axis('off')
plt.suptitle(title)
plt.show()
1. Extract PCA Eigenvectors (Eigenfaces)
PCA components_ shape is (n_components, n_features)
pca_components = pca_std.components_
2. Extract Autoencoder Weights
AE Weights shape is (n_features, n_hidden). We need to transpose to match PCA shape.
ae_weights = encoder_layer.get_weights()[0].T

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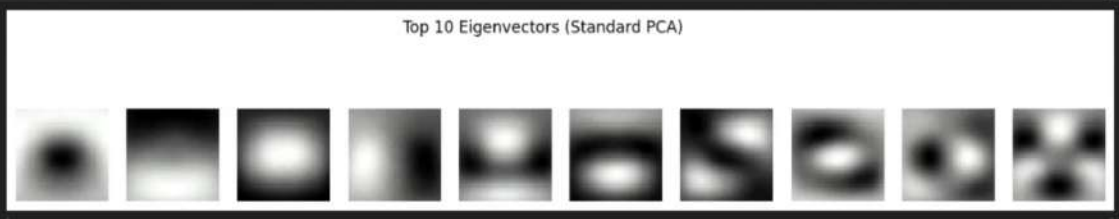
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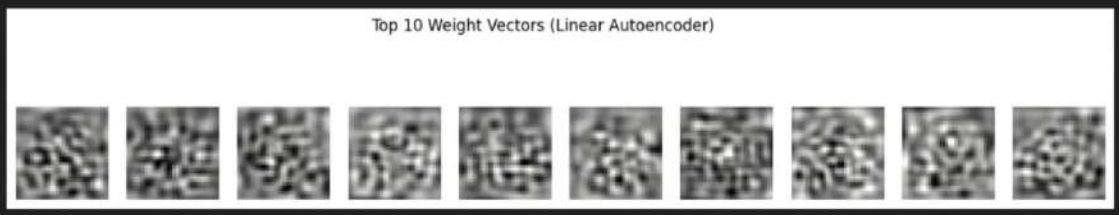
home > cloud > Desktop > adl_assignment1.ipynb > compare eigen vectors from task 1 with autoencoders > # --- Visualization of Components ---

Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ··· base (Python 3.13.9)

```
plot_components(pca_components, "Top 10 Eigenvectors (Standard PCA)")
plot_components(ae_weights, "Top 10 Weight Vectors (Linear Autoencoder)")
```

[9] ✓ 0.2s Python





OUTLINE

TIMELINE

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